

Variation in Swing Path Tilt and Its Relationship to Offensive Performance

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1) Abstract

This project investigates the sources of variation in swing path tilt for the 2025 MLB Season and swing path tilt's relationship to offensive performance and streakiness. The impetus for this project is due to the lack of public research relating MLB's new pitch-by-pitch bat tracking data, specifically swing path tilt, to offensive outcomes. There also hasn't been public research looking into the factors that contribute to variation within these metrics. Studying variation in swing path tilt allows us to understand how swing path tilt can be used to evaluate hitters' approaches and swings. The main hypothesis for this project is that the variation in swing path tilt can be broken down into three factors: timing, mechanical inconsistency and the conscious choice for a batter to change their swing, and noise. Generalized Additive Models (GAMs) are used to find out how much each of the three factors contributes to the overall variation. Roughly 1.5% of the variation can be attributed to timing, 68.4% to mechanical inconsistency and conscious choice, and 30.1% to noise. Mixed-effects models evaluate the relationship between swing path tilt and offensive performance (wOBA) and streakiness (wOBA across 10-game, non-rolling windows). Results show that a very small but positive relationship between swing path tilt and wOBA exists. There does not seem to be any relationship between variance in swing path tilt and offensive streakiness. A linear regression model also measures the relationship between each hitter's strikeout rate in the 2025 season and the correlation between their actual and predicted (based on count, pitch group, and location) swing path tilt. There is a statistically significant (95% confidence level) negative relationship between the correlation of actual and predicted swing path tilt and strikeout rate, implying that hitters whose swing path tilt is closer to what the context of the pitch suggests it should be tend to strike out slightly less.

2) Introduction

To understand a hitter's swing, one of the most informative swing metrics is swing path tilt. Swing path tilt "is defined as the angle of the bat path over the 40 msec prior to contact" (Petriello 3). This metric describes the shape of a hitter's swing by showing whether the bat path is flat or steep (Petriello 3).

My research questions are: "What are the sources of variation in swing path tilt and is there a relationship between swing path tilt and offensive performance among MLB hitters in the 2025 MLB season? How does a hitter's ability to adjust their swing path tilt to pitch context relate to strikeout rate? Does a hitter's variability in their swing path tilt relate to offensive streakiness at all?" The main goal of my project is to determine the factors of variation in swing path tilt and what percentage each factor contributes to overall variation in swing path tilt. Then,

I will look into the relationship between swing path tilt and offensive performance and strikeout rate. I also do research into if a hitter's variation in swing path tilt relates to offensive streakiness at all.

The topic of streakiness has been looked at in the public a lot, but hasn't been so using MLB's new bat metrics and path data, which became public in 2024. The existing public research on streakiness is on if streakiness patterns in baseball occur, and if so, how often. Additionally, there is some public research on what can be learned from the pitch-level swing data, including how hitters can be grouped by their swing characteristics and relationships between the swing metrics. Overall, however, there is no public research about the relationship between metrics about swing characteristics and streakiness. This is another way I want my research to be impactful.

3) Literature Review

Trent McCotter in "Evidence That Hitting Streaks Aren't Just By-Products of Random Variation" found that some hitters exhibited streaks more often than random chance would predict (6). To do this, McCotter analyzed two million hitter batting lines from 1957-2006 (4). He removed all games in which hitters went 0-0 and then in each season randomly reshuffled each player's game log (McCotter 4). This allowed McCotter to get rid of the chronology of the games within each season but keep each player's overall batting average and other statistics for that season. McCotter, with Dr. Peter Mucha of North Carolina's Mathematics department, repeated the randomization 10,000 times for each player-season to determine the expected number of hitting streaks with each of their lengths (4). McCotter and Mucha did this to model streaks as independent events. McCotter and Mucha found that "almost *three times* as many 30-plus game hitting streaks have occurred in real life as we would have expected" for the 1957-2006 MLB seasons (5). Specifically, McCotter found that the expected mean for the number of 5-plus game hitting streaks was 13.5 deviations lower than the actual number of 5-plus game hitting streaks, which could not have occurred by chance (6). McCotter looked into multiple different explanations as to why this is. The most likely explanation he found was that hitters being hot could contribute to these hot streaks occurring more than expected (McCotter 12). This was a more viable explanation than hitters changing their behavior to try and keep up their long hitting streaks (McCotter 12). McCotter did not look into what types of hitters have more streaks than others, which is how my research would build upon McCotter's.

Christian Albright, in "A Statistical Analysis of Hitting Streaks in Baseball," found similar conclusions to McCotter. Albright used several different statistical methods, including logistic regression, to see if "streaks occur more (or less) frequently than would be predicted by a probabilistic model of randomness" (1). He looked at the 1987-1990 MLB seasons and found that, in certain years, players do exhibit streakiness (Albright 1). However, this streakiness was not found across the four years in the study (Albright 1). Albright's work provides more evidence that within season streakiness for hitters is a real phenomenon that occurs more than randomness would suggest.

Joel R. Bock, Akhilesh Maewal, and David A. Gough looked at whether hitting is contagious, i.e., can a hitter's streakiness affect the performance of other hitters in their lineup (1). They found that there is statistical evidence of teammates overall tending to hit better when a hitter in their lineup is in a hot streak (Bock et. al 1). They were not able to determine whether this effect was due to causation, rather than just association (Bock et. al 1). This area of research

is not what I will be focusing on in my project; however, it is important to be aware of the different areas of public research that have been done for offensive streakiness.

Ben Clemens of Fangraphs at the end of August in 2025 started to get closer to my area of research within offensive streakiness. He tried to see if we can tell which types of hitters are streaky (Clemens 1). For the 2021-2024 MLB seasons, Clemens looked at different statistics (ISO, K%, BB%, swing rate, contact rate, chase rate) to see their correlation to streakiness over 20-game stretches in which hitters had at least 40 plate appearances (1-2). The offensive statistic he used to measure performance for streakiness was wOBA, and his baseline was a 50 point deviation from a player's average wOBA over the course of the previous calendar year (Clemens 4). He set a 400 PA minimum for the previous calendar year (Clemens 4). Clemens found the same conclusion as McCotter and Albright did: hot and cold streak stickiness for hitters is real, but is overall small (4). Specifically, he found that $P(\text{Hot}) = 21.7\%$, $P(\text{Cold}) = 20.3\%$, $P(\text{Hot}|\text{Hot}) = 24.8\%$ and $P(\text{Cold}|\text{Cold}) = 24.3\%$ (Clemens 4). Clemens then put many of the different statistics mentioned above into multivariate regressions to predict the stability of offensive streakiness (5). None of the results of the multivariate regressions were statistically significant (Clemens 5). However, Clemens did find that "batters with high walk rates were slightly more likely to have their hot streaks persist, though below a statistically significant level. Batters who hit for a lot of power are less likely to have cold streaks persist, again not at a statistically significant level, but close to it" (5-6). These findings from Clemens start grasping the surface of my research question, and even initially go against my initial hypothesis, which is the following. Overall, power hitters have less variability in their swing path tilt, which I hypothesize would lead to more streakiness. Clemens ultimately concluded that he does not have "much clue about what makes a player streaky or un-streaky" (6). This research done by Clemens is the closest I've found in the public related to my research question, giving me work to build off of.

Jack Lambert, in "Using MLB Bat Tracking Data to Better Understand Swings," concluded that "the top swing accelerations and time to contact values come from generating high bat speed relative to swing length" (6). The correlation between swing acceleration and bat speed was 0.74, between time to contact and swing length was 0.62, between time to contact and bat speed was 0.23, and between swing length acceleration and length was 0.08 (Lambert 6). To determine which types of swing metrics lead to certain offensive outcomes, Lambert grouped hitters using K-Means Clustering (6-7). First, Lambert grouped hitters by bat speed and swing length, creating four clusters, each cluster being one of the quadrants of the bat speed and swing length graph with each data point representing one MLB hitter (7). The most interesting and consequential finding for my project was that the top right quadrant cluster, the hitters with the fastest bat speed and the longest average swing length, had the highest wOBA, barrel percentage, and ISO by a significant amount but also the highest whiff percentage (Lambert 8). These are the hitters that I would hypothesize are more streaky than the rest. Lambert also looked at different clusters created by adding in swing acceleration, time to contact, and attack angle (8,11). Lambert's final conclusion was that "a higher bat speed opens the door to a multitude of different approaches" while lower bat speeds lead to hitters having a narrow margin for error (14). Lambert's work gives me a foundation for how I should think about grouping hitters based on pitch level swing path metrics, if I decide to group hitters.

For the pitch level bat metrics, it is important to note which ones vary a ton pitch-to-pitch, which ones are dependent upon the hitter's timing, and which ones are dependent upon the location of the pitch. Mike Petriello does a great job of defining all of the swing metrics in his article, "4 New Swing Metrics Tell Us More than We've Ever Known about Contact."

Matthew Creally also discusses the swing metrics and some of their limitations in his article, “What Can We Learn from Pitch-Level Swing Shape Data?” Creally points out that attack angle and direction vary a lot pitch to pitch while swing tilt is pretty difficult to change for a hitter (8). Swing path tilt is not as dependent upon a hitter’s timing, but is dependent upon the location of the pitch (Petriello 5).

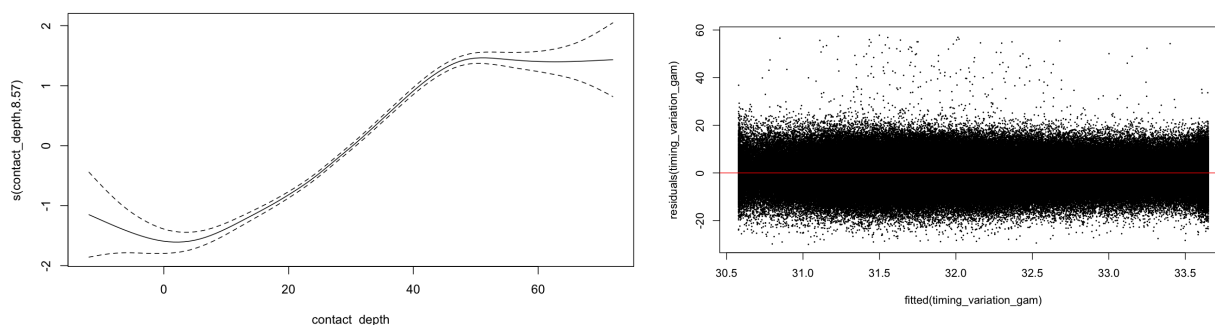
4) Methodology, Results, and Discussion

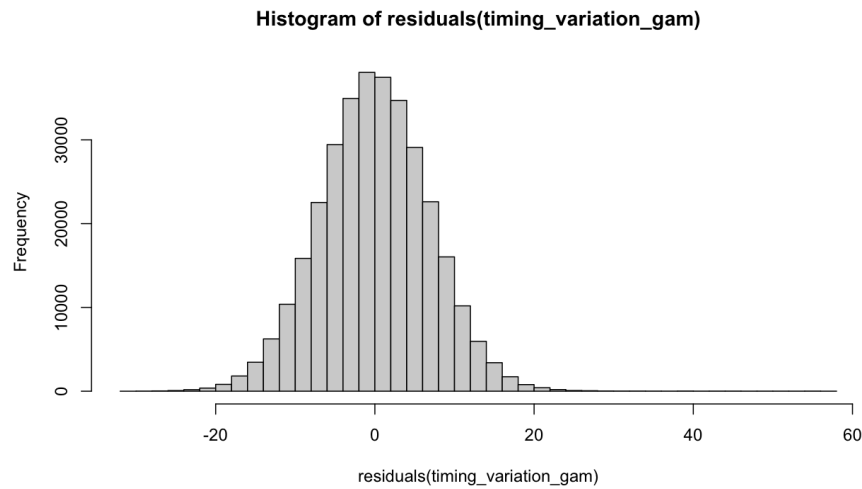
In this project, I used MLB pitch-by-pitch data from Statcast for the 2025 regular season. I defined a swing as a swinging strike, foul ball, or a ball hit into play. I did NOT count a bunt as a swing. This is because swing path tilt is irrelevant for bunts for the purpose of my project. I filtered out NAs for the following variables: batter_id, swing_path_tilt, and contact_depth.

The sources of variation in swing path tilt can be broken up in three ways: (1) **timing**, (2) **conscious choice/mechanical inconsistency (cc/mi)** for a batter, (3) **noise**. The first part of my methodology was figuring out this pie chart of swing path tilt variation. To do this, I used Generalized Additive Models (GAMs). My first GAM is the timing variation gam:

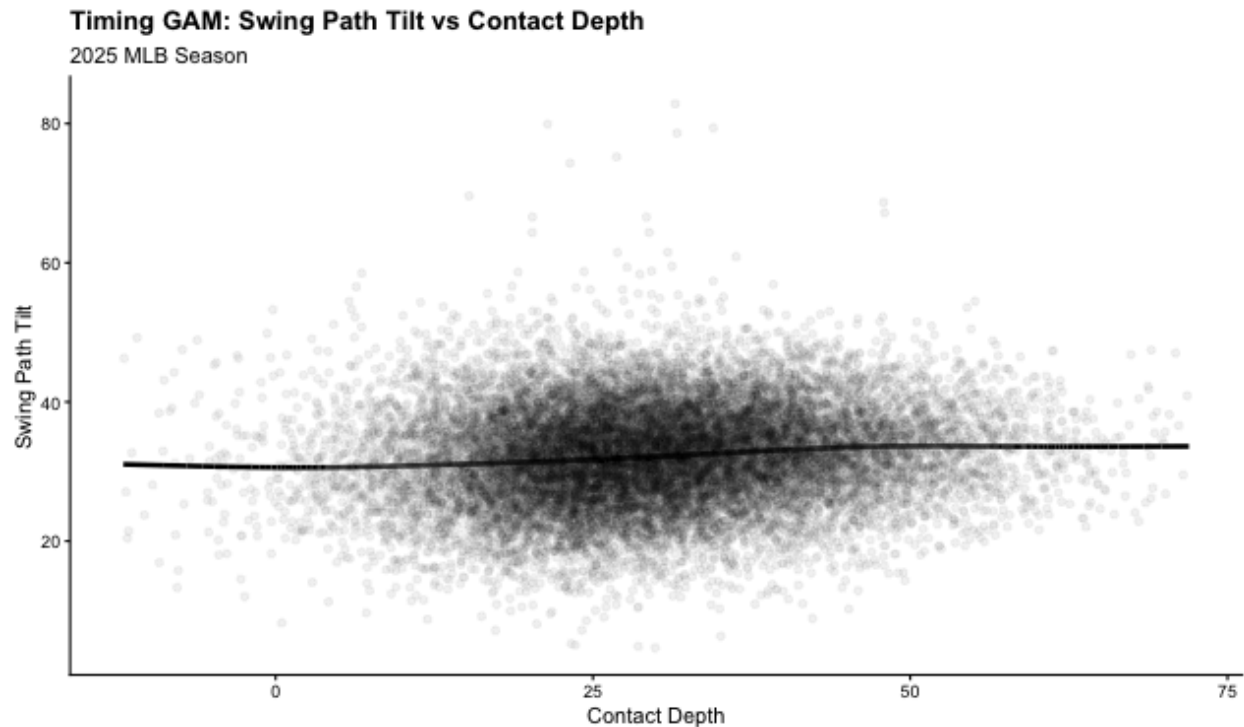
```
gam(swing_tilt ~ s(contact_depth, k = 20), data = swings, method = "REML").
```

I wanted to isolate how much variation in swing path tilt is due to **timing** and see if contact depth alone can explain variation in swing path tilt at the population level. Contact depth measures a hitter’s timing because it quantifies where a hitter made contact with the ball along its path (for a swing and miss, it is the point where the bat is closest to the ball). Contact depth identifies the point in the plane perpendicular to the ground where the batter made contact with the ball. Thus, contact depth can determine whether a hitter was early for a pitch, on time, or late. For this GAM, I analyzed 327,143 swings. I chose $k=20$ for this GAM to allow the smooth term for contact depth to be flexible in order to capture the nonlinear relationship between contact depth and swing path tilt. I ran `gam.check()` on my model to ensure that my choice of k was appropriate. It was, as the edf (effective degrees of freedom) was 8.57. Also, the k -index of 1 had a p -value of 0.63, meaning that my choice of $k=20$ is large enough. Below are graphs that show that my model choice of a GAM was appropriate for my variables of swing path tilt and contact depth:





My F-statistic was 457.8, my p-value was below 0.01, and the GAM had an adjusted R^2 of 0.0147. At the population level, there is a **statistically significant** nonlinear relationship between swing path tilt and contact depth. However, the effect of this relationship is small, which can be seen in the graph below:

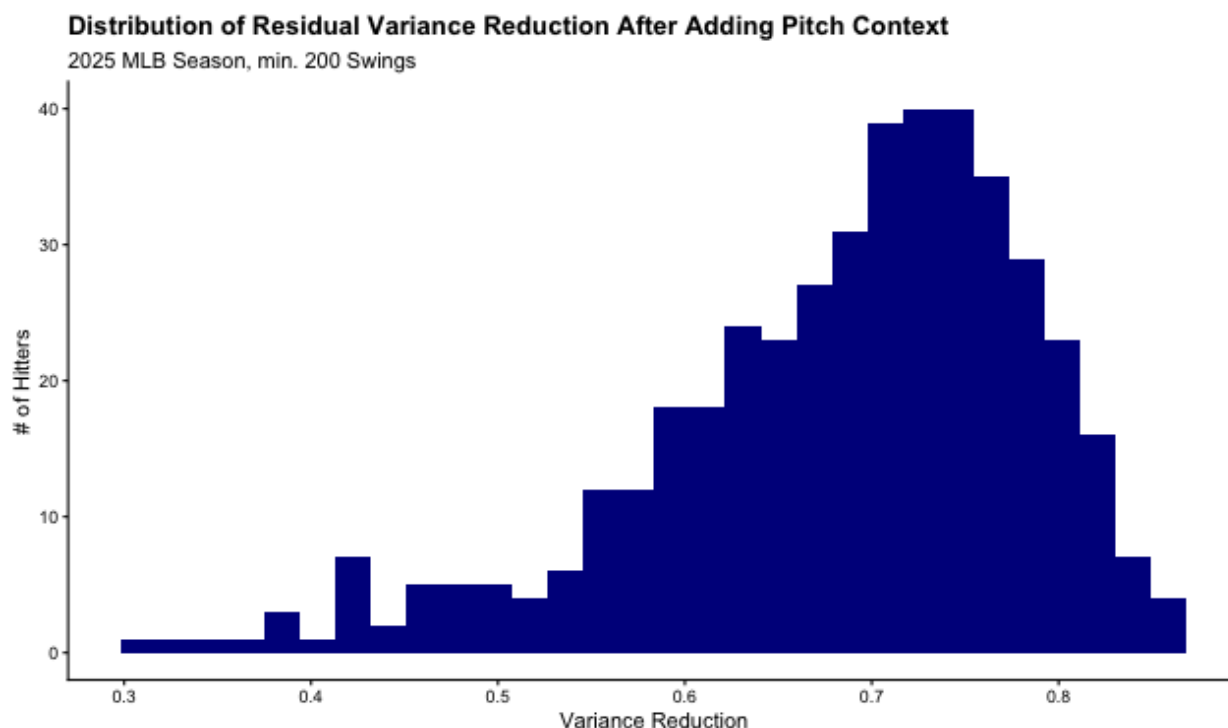


The adjusted R^2 of 0.0147 means that timing (contact depth) explains only 1.47% of the variation in swing path tilt overall.

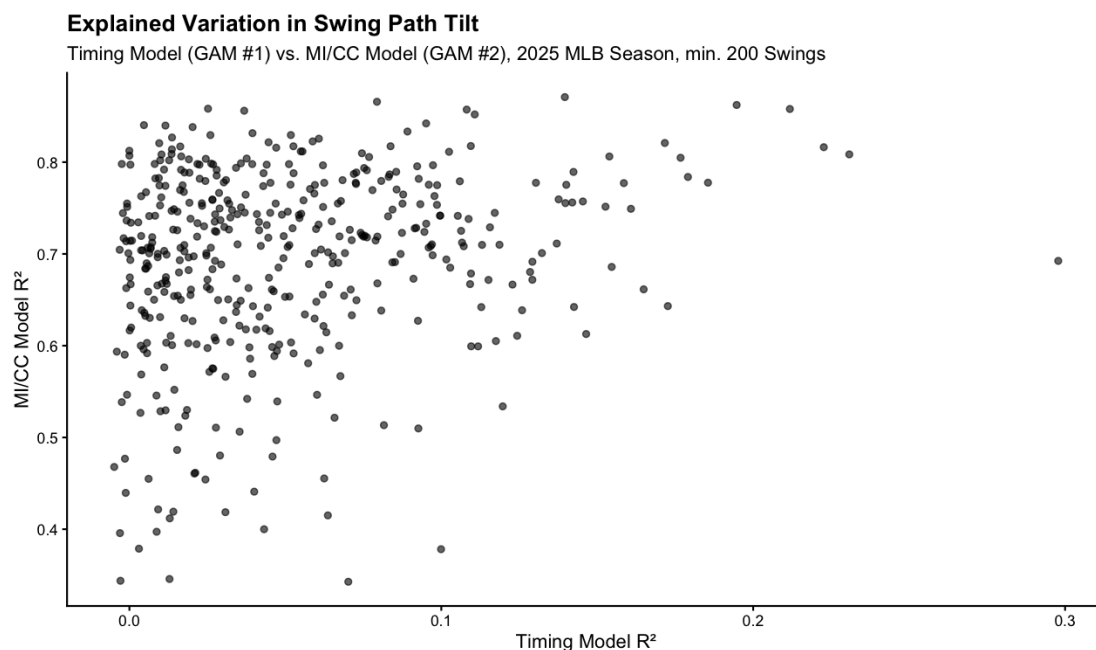
The next GAMs I ran were at the batter level. I looked at all batters in the 2025 MLB season with at least 200 swings. The first GAM I ran at the batter level was the timing only model, with the same formula as the population level except it ran for each batter with $k = 10$. Again, this model explains how much contact depth can explain the variation in each given hitter's swing path tilt. I found that the mean residual variance was 31.5 with a mean R^2 of 0.0505. Thus, for the average hitter, contact depth by itself explains a very small amount, roughly 5%, of the variation in swing tilt. The second GAM I ran added important context for each pitch: pitch grouping (fastball, breaking ball, offspeed, knuckleball), count (balls and strikes), and pitch location, both horizontal and vertical. For this model, the formula was as follows:

$s(\text{contact_depth}, k = 10) + \text{pitch_group} + \text{balls} + \text{strikes} + s(\text{plate_x}, k = 6) + s(\text{plate_z}, k = 6)$.

Now, we can compare the residual in the timing hitter model and this timing + additional context model to see how much the additional context, the **conscious choice/mechanical inconsistency** part of the pie chart, can explain variation in swing path tilt. After adding in these three contextual variables, I found that the mean residual variance was 9.83 and the mean R^2 was 0.6931. Thus, swing path tilt for each hitter is explained much more through contextual factors than just by timing alone. From adding these contextual controls, the mean variance reduction is 0.6836. Therefore, for the average hitter, roughly 68.4% of the residual variance in swing path tilt from the timing GAM is removed after factoring in pitch group, count, and pitch location. The distribution below shows how much each hitter's unexplained variation in swing path tilt after just the timing GAM is removed once pitch group, count, and location are factored in.



The better performance of the model with the three contextual factors compared to the contact depth only model can also be seen in the graph below. In this graph, each point is one hitter, with the timing model on the x-axis and the timing + cc/mi model on the y-axis. The higher a point is on the y-axis than far it is on the x-axis represents the timing + cc/mi model outperforming the timing only model.



In the graph, you can see that for nearly every hitter, the timing + cc/mi model explains more of the variation in swing path tilt than just the timing model. There are two main takeaways from these GAMs. First, swing tilt is very context dependent, as the model fit improves considerably when factoring in pitch group, count, and pitch location. Second, a lot of the variation in swing path tilt is not random. ~30% of the variation in swing path tilt is **noise**, but ~70% is not. This is a significant finding. Studying this variation in swing path tilt is a worthy cause, as it exists because of the factors that are within a team's control (when they are pitching).

Next, I used three linear mixed-effects models to see whether swing path tilt is associated with wOBA for swings for the 2025 MLB season that ended a plate appearance (PA). I chose wOBA as the dependent variable rather than xwOBA, which is a more predictive metric, because I wanted to reward hitters who change their swing path in certain situations to get a hit or “do their job.” For example, think of a RHH who is up at the plate with a runner at 3B, two outs, and has an 0-2 count. The pitcher throws a slider away. This hitter, to try and get the runner in, changes their swing and pokes a single into RF, scoring the run. The exit velocity on the hit could be 78 mph, but the hitter did their job and got an RBI with two outs and a runner in scoring position. I used all swings from the 2025 MLB season with non-NA/missing batter_id, pitcher_id, contact_depth, swing_tilt, and woba_value. The first linear mixed-effects model was my baseline model. This model tested whether swing path tilt is associated with wOBA while including random effects for each batter and pitcher. The formula is:

$$\text{woba_value} \sim \text{swing_tilt} + (1 \mid \text{batter_id}) + (1 \mid \text{pitcher_id}).$$

I found that swing path tilt does statistically predict wOBA at a significant level with a p-value below 0.01. Furthermore, there is a positive yet extremely small relationship between swing path tilt and wOBA. An increase in swing path tilt by ten degrees corresponds to an 0.0114 increase in wOBA. The second model follows a similar framework to the batter GAMs. It is a mixed effects model with pitch group, contact depth, count (balls and strikes), pitch location, batter handedness, pitcher handedness, and batter and pitcher random effects included. The exact formula is:

$$\text{woba_value} \sim \text{swing_tilt} + \text{contact_depth} + \text{pitch_group} + \text{balls} + \text{strikes} + \text{plate_x} + \text{plate_z} + \text{stand} + \text{p_throws} + (1 \mid \text{batter_id}) + (1 \mid \text{pitcher_id}).$$

The main question this model answers is: Does swing path tilt predict wOBA after controlling for the factors above? Similar to the GAMs, the mixed effects model fit improved a lot. The AIC dropped by ~4,014 and pitcher and residual variance decreased. Furthermore, an increase in swing path tilt by ten degrees with this model corresponds to an 0.0393 increase in wOBA. This was statistically significant with a p-value of 0.00177. Other intuitive findings from the model are that balls increase wOBA and higher vertical plate location leads to an increase in wOBA. The third mixed effects model I ran is a random-slope model. The formula is:

$$\text{woba_value} \sim \text{swing_tilt} + \text{contact_depth} + \text{pitch_group} + \text{balls} + \text{strikes} + \text{plate_x} + \text{plate_z} + \text{stand} + \text{p_throws} + (1 + \text{swing_tilt} \mid \text{batter_id}) + (1 \mid \text{pitcher_id}).$$

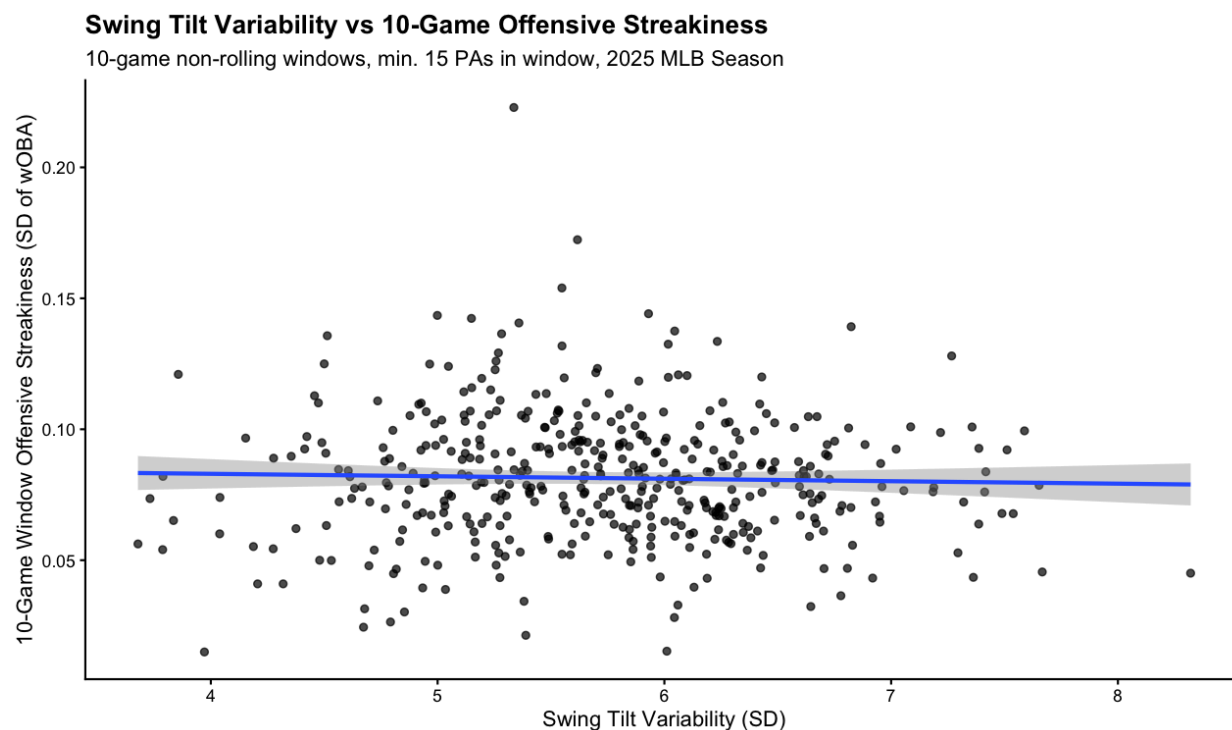
The one difference compared to the second mixed effects model is the addition of the slope for swing tilt for each unique batter. The goal of this model is to see how the effect of swing path tilt on wOBA is different across hitters. This random-slope mixed effects model fits the data just slightly better than the second mixed effects model above. The variance for the slope is 0.00000284, implying that the effect that swing tilt has on offensive outcomes does not change much across hitters.

Overall, from these three models, I found that in all three a positive relationship between swing path tilt and wOBA exists. The relationship is stronger after controlling for the context of the pitch. This makes the second mixed-effects model, the one with these controls but no batter random-slope, the best one. The interpretability and stronger fit of this model than the baseline model contributes to this.

Then, I used the three mixed effects models above to see whether there is a relationship between a hitter's variation in their swing path tilt and wOBA in non-rolling, non-overlapping 10-game windows (with at least 15 PAs) in the 2025 MLB season. I looked at batters who had at least 3 of these windows. Thus, I am testing the relationship between swing path tilt variance and streakiness in wOBA. My hypothesis for this part of the project is that hitters with less variance in their swing path tilt are more streaky. The formula for the first, baseline model I ran was:

$$\text{sd_window_woba} \sim \text{sd_swing_tilt}.$$

I found that there is no relationship between the standard deviation of wOBA and the standard deviation of swing path tilt in each 10-game window. The R^2 value was 0.0009. Thus, variance in swing tilt on its own did not explain any of the variance in wOBA in 10-game windows. This can be seen in the graph below, where each point is one batter who had at least 3 10-game windows. The x-axis is each hitter's swing path tilt variability and the y-axis is each hitter's standard deviation of offensive streakiness, as measured by wOBA.



The graph shows that there is no relationship between variability in swing path tilt and offensive streakiness in 10-game windows for hitters in the 2025 MLB season. For the second model, my formula was:

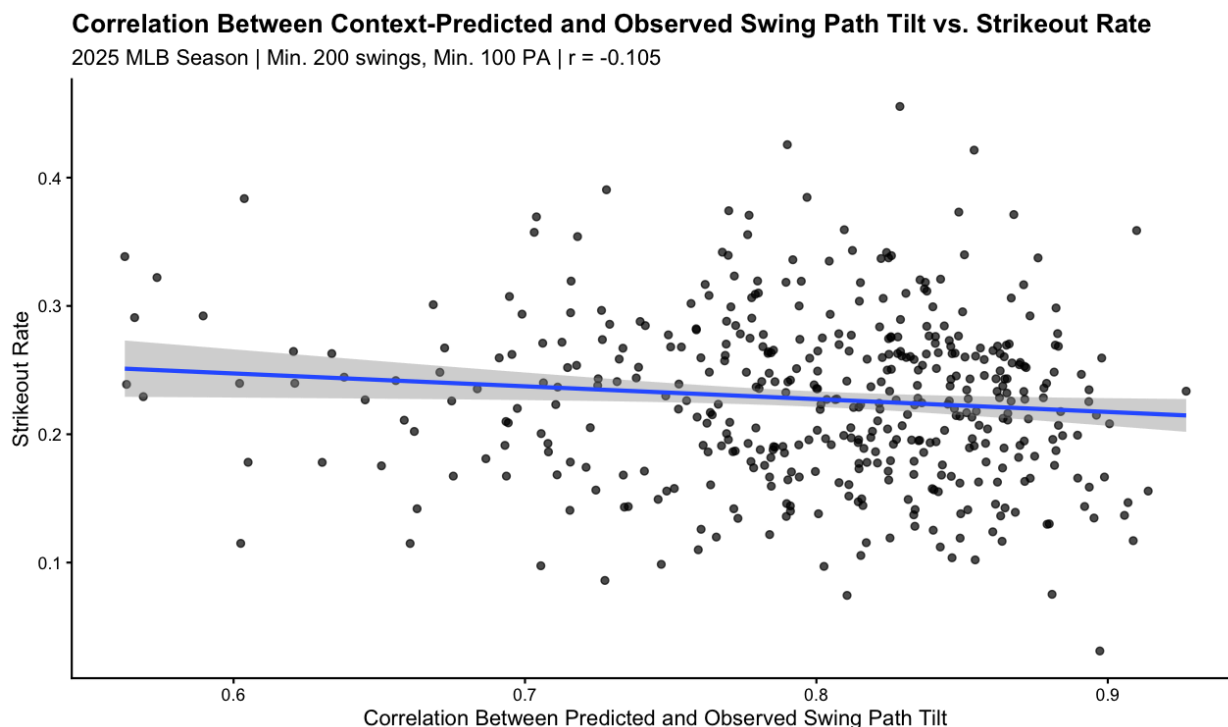
$$\text{sd_window_woba} \sim \text{sd_swing_tilt} + \text{mean_window_woba} + \text{number_10_game_windows} + \text{number_swings}.$$

I controlled for each hitter's average wOBA, the number of 10-game windows they had, and their overall number of swings in the 2025 season. Just like the first model, there is no relationship between variability in swing path tilt and offensive streakiness in 10-game windows, as the second model has an R^2 of 0.0337. For the third model, I used the coefficient of variation for the wOBA and swing path tilt in each window. The formula I used is:

$\text{coefficient_of_variation_window_woba} \sim \text{coefficient_of_variation_swing_tilt} + \text{mean_window_woba} + \text{number_windows_10} + \text{number_swings}.$

The R^2 of 0.0718 was slightly higher than the second model. The p-value was 0.649. Thus, there is no relationship between relative swing tilt variability and relative offensive streakiness in 10-game windows for hitters in the 2025 MLB season. Therefore, my initial hypothesis that hitters with less swing tilt variability were more streaky was incorrect.

Lastly, I used a GAM to investigate the relationship between swing path tilt and strikeout rate. I looked at all hitters in the 2025 MLB season who had at least 200 swings and 100 PA. I filtered out automatic balls, pitchouts, and eephus pitches. I fit a GAM to predict swing path tilt based on pitch group, pitch location, and count. This is the MI/CC GAM, but without contact depth. I did not include contact depth in this GAM because contact depth is a result of the swing. Then, I calculated the correlation between every hitter's observed swing path tilt and their predicted swing path tilt from the GAM. This correlation represents how consistently a hitter's swing path tilt adapted to the context of the pitch. I measured the correlation between each hitter's strikeout rate and the correlation between their observed vs. predicted swing path tilt. This correlation was -0.105, indicating that hitters whose swing path tilt was closer to the predicted swing path tilt based on count, pitch group, and pitch location tended to have slightly lower strikeout rates. In the graph below, the blue line is the linear regression trend line, showing this negative relationship. The slope is -0.10003, which implies that a 0.1 increase in the correlation between predicted and observed swing path tilt is associated with a 1 percentage point decrease in strikeout rate.



5) Conclusion

In this project, I used GAMs to investigate the sources of variation in swing path tilt. I found that contact depth explains 1.5% of the total variation in swing path tilt. When adding in count, pitch group, and pitch location, 68.4% of the total variation can be explained. I then used linear mixed-effects models to look at the relationship between swing path tilt and wOBA and found that there is an extremely small but positive relationship. There does not appear to be any relationship between a hitter's variation in their swing path tilt and offensive streakiness. I measured streakiness through a hitter's wOBA across 10-game, non-rolling windows. Finally, I ran a GAM to predict a hitter's swing path tilt based on count, pitch group, and pitch location. I found that hitters whose swing path tilt more closely aligns with what the count, pitch group, and pitch location suggest it should be tend to strike out slightly less. One limitation of this project is that I used only 2025 data. As more bat tracking data becomes available to the public, it would be beneficial to continue looking into the sources of variation in swing path tilt and swing path tilt's relationship to offensive performance. Future research on this topic could include analysis of the relationship between swing path tilt and other offensive metrics and statistics.

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